

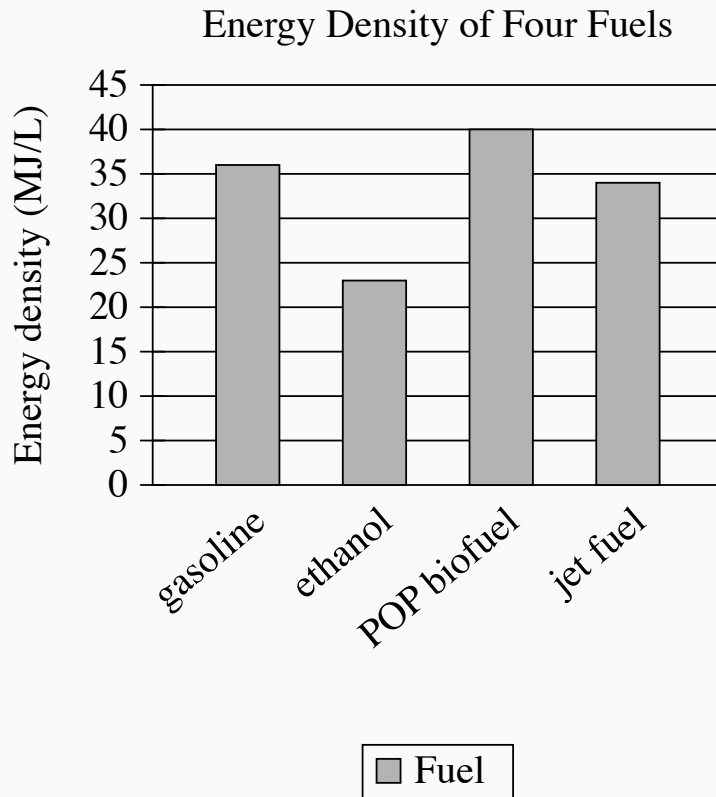
● EASY

● INFORMATION AND IDEAS

Information and Ideas

04

10 questions · ~15 min



- The Energy density data for the 4 bars are as follows:
 - gasoline: 36 MJ/L
 - ethanol: 23 MJ/L
 - POP biofuel: 40 MJ/L
 - jet fuel: 34 MJ/L

A team of researchers used bacteria to create a biofuel (a renewable fuel produced from plants, algae, or other living materials). The team called the new fuel POP biofuel. To determine how much energy is stored in POP biofuel compared to other fuels, the team calculated the fuels' energy density, in megajoules per liter (MJ/L). The team found that the fuel with the highest energy density is _____blank

Which choice most effectively uses data from the graph to complete the sentence?

- A. jet fuel.
- B. POP biofuel.
- C. ethanol.
- D. gasoline.

A typical string quartet consists of two violin players, a viola player, and a cello player. When a string quartet performs, one violinist and the cellist usually sit facing the audience. The violist and the other violinist usually sit facing each other. This seating arrangement causes the viola's carved sound openings, or f-holes, to be angled away from the audience. Therefore, if the viola player has a musical solo and wants to make sure that the audience can hear it clearly, the violist typically _____ blank

Which choice most logically completes the text?

- A. instructs the other members of the quartet to play louder.
- B. plays at the same volume as the cellist.
- C. shifts position in order to face outwards toward the audience.
- D. looks at the sheet music instead of looking at the audience.

In 2022, researchers rediscovered ancient indigenous glyphs, or drawings, on the walls of a cave in Alabama. The cave's ceiling was only a few feet high, affording no position from which the glyphs, being as wide as ten feet, could be viewed or photographed in their entirety. However, the researchers used a technique called photogrammetry to assemble numerous photos of the walls into a 3D model. They then worked with representatives of tribes originally from the region, including the Chickasaw Nation, to understand the significance of the animal and humanoid figures adorning the cave.

According to the text, what challenge did the researchers have to overcome to examine the glyphs?

- A. The cave was so remote that the researchers couldn't easily reach it.
- B. Some of the glyphs were so faint that they couldn't be photographed.
- C. The researchers were unable to create a 3D model of the cave.
- D. The cave's dimensions prevented the researchers from fully viewing the glyphs.

Average Prices Received by US Growers for Citrus Fruits, 2020–2021
(dollars per box)

Fruit	June 2020	June 2021	July 2020	July 2021
Grapefruits	\$13.80	\$23.72	\$16.13	\$22.98
Oranges	\$16.15	\$11.09	\$15.53	\$13.79
Lemons	\$19.50	\$28.94	\$21.01	\$31.78

An employee of a citrus grower in the United States is analyzing the prices of several varieties of fruit over the course of a growing season. The employee wishes to know how much a grower could have expected to receive on average for a box of lemons in July 2021. Consulting the table, the employee finds that this amount is _____blank

Which choice most effectively uses data from the table to complete the statement?

- A. \$31.78.
- B. \$23.72.
- C. \$22.98.
- D. \$28.94.

Microbes that live in shallow lakes and ponds produce methane, a harmful greenhouse gas. Ecologist Ralf Aben and his team wanted to see how different types of shallow-water plants might affect the amount of methane that escapes into the atmosphere. Aben's team set up some water tanks with soil and microbes from local ponds. Some tanks had a type of underwater plant that grows in the soil called watermilfoil. Other tanks had either duckweed, a type of plant that floats on the water's surface, or algae. Aben and his team found that tanks with duckweed and algae released higher levels of methane than tanks with watermilfoil did. This finding suggests that _____ blank

Which choice most logically completes the text?

- A. the presence of some kinds of underwater plants like watermilfoil helps prevent methane from escaping shallow lakes and ponds.
- B. shallow lakes and ponds release more methane than deeper bodies of water because shallow bodies of water usually have more plants than deep bodies of water do.
- C. shallow lakes and ponds are more likely to contain algae than to contain either watermilfoil or duckweed.
- D. having a mix of algae, underwater plants, and floating plants is the best way to reduce the amount of methane in shallow lakes and ponds.

The following text is from Shyam Selvadurai's 1994 novel *Funny Boy*. The seven-year-old narrator lives with his family in Sri Lanka. Radha Aunty is the narrator's aunt.

Radha Aunty, who was the youngest in my father's family, had left for America four years ago when I was three, and I could not remember what she looked like. I went into the corridor to look at the family photographs that were hung there. But all the pictures were old ones, taken when Radha Aunty was a baby or young girl. Try as I might, I couldn't get an idea of what she looked like now. My imagination, however, was quick to fill in this void.

©1994 by Shyam Selvadurai.

According to the text, why does the narrator consult some family photographs?

- A. He wants to use the photographs as inspiration for a story he is writing.
- B. He is curious about how his father dressed a long time ago.
- C. He hopes the photographs will help him recall what his aunt looked like.
- D. He wants to remind his aunt of an event that is shown in an old photograph.

In addition to her technical skill and daring feats, American stunt pilot Bessie Coleman was also known for dazzling the crowds that came to watch her air shows in the 1920s with her exuberant personality. During her career, she was careful and purposeful about how she crafted her public persona. An aviation researcher has claimed that Coleman intentionally defied social norms of the time by how she chose to present herself to the public.

Which quotation from an article about Coleman would most directly support the aviation researcher's claim?

- A. "For her air shows, Coleman frequently used the Curtiss JN-4, or 'Jenny,' which at that time was one of the most well-known types of planes."
- B. "While Coleman was beloved by spectators for her charisma, she had a more complicated relationship with her managers and staff, who at times found her behavior too impulsive and demanding."
- C. "Coleman once considered leaving her career as a stunt pilot to focus her efforts on giving speeches, which she felt would better support her public image."
- D. "Although female pilots were typically expected to wear traditional but impractical attire that included dresses or skirts, photographs of Coleman show her wearing pants and leather jackets."

The Nacional tree is a rare variety of cacao. Nacionales were thought to have gone extinct by the twentieth century due to a fungus. This fungus can spread from tree to nearby tree through the air and causes disease. But around 2013, cacao expert Servio Pachard located some of these Nacional trees. The trees were in the Piedra de Plata coastal forest, within a hard-to-reach valley in Ecuador. Conservationists inferred that the Nacional trees in Piedra de Plata might have avoided the diseases that wiped out the other Nacionales because _____ blank

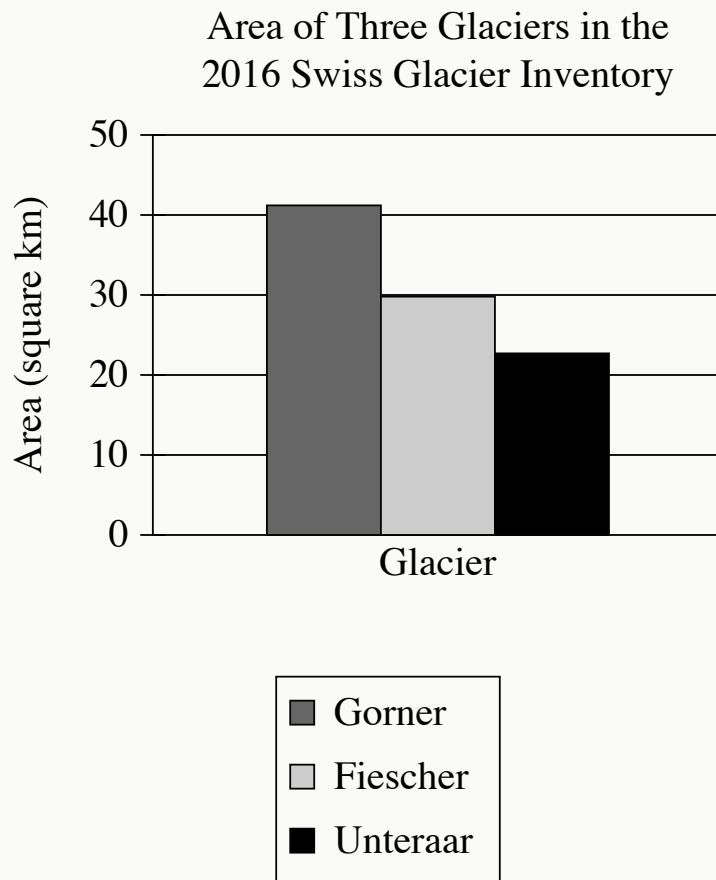
Which choice most logically completes the text?

- A. early twentieth-century scientists did not know why so many Nacionales were becoming infected.
- B. the ability of the fungus to travel through the air was only recently discovered.
- C. they were too far from the other Nacional trees infected by the fungus to become infected themselves.
- D. the chocolate made from their pods was highly valued.

Scent is tightly interwoven with our daily lives, often evoking significant memories and important social events. This connection is of growing interest to archaeologists who hope to use it to better understand ancient rituals, trade, social hierarchies, and medicine. Although the speed at which odor molecules dissipate makes identifying ancient scents challenging, advancements in biomolecular technologies show promise in unlocking ancient aromas from preserved artifacts. Archaeological studies making use of these advancements may provide new insights into past societies.

According to the text, what is one reason some archaeologists are interested in recovering scents from ancient artifacts?

- A. They are investigating whether people's sense of smell has declined in recent centuries.
- B. They believe the scents could illuminate important aspects of ancient life.
- C. They think that ancient scents would be enjoyable to people today.
- D. They hope to develop new medicines using ancient scent molecules.



- The data for the 3 categories are as follows:
 - Gorner: 41.2 square kilometers
 - Fiescher: 29.8 square kilometers
 - Unteraar: 22.7 square kilometers

To monitor changes to glaciers in Switzerland, the government periodically measures them for features like total area of ice and mean ice thickness, which are then reported in the Swiss Glacier Inventory. These measurements can be used to compare the glaciers. For example, the Gorner glacier had _____ blank

Which choice most effectively uses data from the graph to complete the example?

- A. a larger area than either the Fiescher glacier or the Unteraar glacier.
- B. a smaller area than the Fiescher glacier but a larger area than the Unteraar glacier.
- C. a smaller area than either the Fiescher glacier or the Unteraar glacier.
- D. a larger area than the Fiescher glacier but a smaller area than the Unteraar glacier.